

SatQuery AI - Multimodal Remote Sensing Evidence Report

Autonomous Agentic Execution & Evidence Provenance Trace

Query:	"What changed between these two dates, and where did the change occur?"
Task Category:	CHANGE_VQA
Specialist Tool:	Bi-Temporal Change Understanding & CDVQA Specialist
Reliability Score:	94.9% (HIGH TRUST)
Execution Time:	55.1 ms

1. Agent Synthesis & Visual-Language Answer

Bi-Temporal Multi-Image Change Analysis & CDVQA Output:

Bi-temporal change analysis identifies significant land-cover conversion between Observation Date T1 and Date T2.

- 1. Built-Up / Developed Area: HAS INCREASED by 16.51% across the scene tile (43,285 pixels).
- 2. Primary Hotspot: South-Eastern quadrant.
- 3. Conversion Profile: Former vegetated/natural land transitioned into new building structures, roofs, and impervious pavement.

Spatial Change Map generated with contour boundary tracking and pixel-level diff.

2. Auditable Agent Pipeline Execution Trace

- **[Perception Layer]** Format & CRS validated. Mode: Bi-temporal T1 + T2 (2 image(s)). Status: PASSED
- **[Query Engine]** Classified intent: 'CHANGE_VQA'. Extracted entities: ['general_land_cover']. Task: 'change_vqa'
- **[Mission Planner]** Selected tool: 'Bi-Temporal Change Understanding & CDVQA Specialist' with params {'threshold': 35, 'blur_kernel': [5, 5], 'contour_min_area': 100}. Generated 7 node task-graph.
- **[Specialist Tool Execution]** Executed specialist tool 'Bi-Temporal Change Understanding & CDVQA Specialist' with model confidence 94.5%.
- **[Evidence Fusion]** Aligned 3 evidence streams. Temporal Alignment Confirmed: Spatial change map cross-validated across T1-T2 frames.
- **[Trust & Uncertainty]** Calculated Reliability Score: 94.9% (HIGH TRUST). Cross-model agreement: 94.5%.
- **[Discovery Engine]** Autonomous scan completed: Autonomous scan identified 2 'Beyond Your Query' discovery item(s).
- **[Answer Generator]** Final evidence-linked response synthesized in 55.1 ms.